HANWEN FAN

Ph.D. Student in Mechanical Engineering | Computational and Experimental Biomechanics

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Ph.D. researcher studying multiscale bone mechanics and regeneration by integrating full-field experiments, 3D imaging, image-based finite element analysis, scientific machine learning, and biomaterials characterization.

RESEARCH INTERESTS

Computational biomechanics; multiscale bone mechanics; image-based FEA; biomaterials characterization; regenerative biomechanics

EDUCATION

Texas A&M University, Ph.D. in Mechanical Engineering Expected May 2028
GPA: 3.80/4.00

University of Texas at Dallas, M.S. in Mechanical Engineering May 2024
GPA: 3.97/4.00

Xi'an Jiaotong University, M.S. in Power Engineering 2021

Xi'an Jiaotong University, B.S. in Energy and Power Engineering 2018

RESEARCH EXPERIENCE

Texas A&M University, Research Assistant, Mechanical Engineering Jun 2024 - Present
Advisor: Prof. Yuxiao Zhou | Computational and experimental biomechanics

- Develop an inverse-mechanics workflow that combines 3D DVC displacement fields, PyTorch, spatial modulus reconstruction, and heterogeneous Abaqus models for full-field material-property estimation and validation.
- Integrate compression testing, micro-CT, DVC, MRI-derived composition, and image analysis to quantify local strain and microstructural drivers across osteolytic and osteoblastic metastatic lesions in human trabecular bone.
- Build micro-CT-derived finite element models and analyze trabecular thickness and connectivity to isolate structural contributions to metastatic bone strength.
- Link whole-bone strain mapping with AFM nanomechanical mapping, Raman spectroscopy, and AFM-IR to characterize multiscale stiffness and compositional heterogeneity in osteogenesis imperfecta bone.
- Develop micro-CT-based Abaqus models of mouse digit-tip regeneration to examine how evolving geometry and mechanical stimuli coordinate tissue regeneration.

University of Texas at Dallas, Research Assistant, Nano-Thermal Bioengineering Lab Aug 2021 - May 2024
Advisor: Prof. Zhenpeng Qin | Nanomedicine and blood-brain barrier transport

- Engineered antibody-conjugated gold nanoparticles and conducted in vivo laser-mediated blood-brain barrier modulation studies using cell culture, small-animal procedures, histology, and fluorescence imaging.
- Developed quantitative ImageJ workflows to evaluate imaging and drug-delivery outcomes for targeted therapeutic transport.

AWARDS AND RESEARCH SUPPORT

TMS Biomaterials Best Student Poster Award, 2nd Place 2026
Biomechanics in Digit Tip Regeneration

TMS Biomaterials Best Student Poster Award, 2nd Place 2025
Microstructures in Metastatic Bone Lesions Using AFM Nanomechanical Testing

MEEN Graduate Summer Research Grant 2025 and 2026

TEACHING EXPERIENCE

Texas A&M University, Teaching Assistant, Solid Mechanics Aug 2024 - Dec 2024

- Led recitation sections and office hours and developed quizzes and exam problems

TECHNICAL EXPERTISE

Programming and machine learning: Python, MATLAB, PyTorch, C++

Modeling and imaging: Abaqus, finite element analysis, computational fluid dynamics, micro-CT, digital volume correlation, confocal imaging, Avizo, ImageJ

Experimental and materials characterization: Mechanical testing, AFM nanomechanical mapping, Raman spectroscopy, AFM-IR, SEM, immunohistology, cell culture

PUBLICATIONS

1. **H. Fan**, P. Nguyen, Z. Üzüm, et al., L. Dawson, C. P. Dolan, Y. Zhou, Multiscale analysis reveals distinct temporal trajectories of morphological reconstruction and mechanical maturation during mouse digit tip regeneration, (2026), Acta Biomaterialia, DOI: 10.1016/j.actbio.2026.08.032
2. H. Lin, S. Xu, P. Xuan, Y. Wang, R. Ge, **H. Fan**, B. F. Stephens, Y. Zhou, X. Dong, Coordinated control of miniature magnetic pumps for liquid sampling and delivery in a knee implant, (2026), Device, DOI: 10.1016/j.device.2026.101203
3. J. Chen, **H. Fan**, X. Xiao, V. Nguyen, Y. Zhou, Multiscale mechanical heterogeneity and structural gradients in the annulus fibrosus-endplate interface in the spine characterized using AFM nanomechanical testing, (2026), Journal of Biomechanics, DOI: 10.1016/j.jbiomech.2026.113383
4. T. Komori, M. Nagata, N. Praneetpong, **H. Fan**, Y. Zhou, N. Ono, W. Ono, Wnt-dependent ontogeny of acellular cementum-forming cementoblasts on the tooth root surface, (2026), Nature Communications, DOI: 10.1038/s41467-026-72712-1
5. Y. Wang, M. Kim, **H. Fan**, R. Ge, D. W. Wang, F. Maldonado, C. T. Demarest, Y. Zhou, X. Dong, A miniature magnetic switch unlocking multimodal, chip-free, and batteryless airway sensing, (2026), Science Advances, DOI: 10.1126/sciadv.aed3998
6. **H. Fan**, E. Busschers, H. Tanoto, et al., Y. Bae, B. Lee, Y. Zhou, Mechanical characterization of Col1a1+/- osteogenesis imperfecta bone revealed altered mechanical stiffness heterogeneity across scales, (2026), Cell Biomaterials, DOI: 10.1016/j.celbio.2025.100246
7. J. Kehl, **H. Fan**, X. Xiao, V. Hoefler, M. Saha, N. Ono, Y. Zhou, Spatiotemporal mechanical stimuli from dental crypt during tooth development: An integrated experimental and computational study, (2026), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2026.102449
8. R. Ge, Y. Wang, C. Negron, **H. Fan**, F. Maldonado, C. Demarest, V. Simon, Y. Zhou, X. Dong, A wireless implantable sensory ring for continuous airway stent migration tracking, (2026), npj Flexible Electronics, DOI: 10.1038/s41528-025-00526-0
9. C.C. Tsai, **H. Fan**, Y. Zhou, S. Xie, Rational design of ionomer microstructures for thermally reprocessable materials with creep resistance and recoverability, (2025), JACS Au, DOI: 10.1021/jacsau.5c01317
10. Y. Zhu, B. Xiao, A. Kirk-Jadric, **H. Fan**, R. Naik, Y. Zhou, X. Dong, Magnetic soft robotic valve for minimally invasive therapy of gastroesophageal reflux disease, (2025), Device, DOI: 10.1016/j.device.2025.100932
11. H. Tanoto, **H. Fan**, J. Z. Chen, C. B. Rodriguez, E. Milton, F. Espinoza, G. Aguilar, C. P. Dolan, N. Ono, Y. Zhou, Investigating simultaneous mineralization across layers during tooth development using atomic force microscopy and Raman spectroscopy, (2025), Journal of the Mechanical Behavior of Biomedical Materials, DOI: 10.1016/j.jmbbm.2025.107094
12. **H. Fan**, Z. Xu, C. Rodriguez, N. Dover, A. Demkov, M. Lilly, G. Aguilar, L. Suva, X. Zhang, Y. Zhou, The microstructure of metastatic bone lesions suggests tumor mediated alterations in bone mineralization, (2025), Bone, DOI: 10.1016/j.bone.2025.117625
13. P.T. Lin, C. Zhang, Y. Tang, **H. Fan**, et al., A. Armstrong, J. Du, Y. Zhou, Full-field strain distribution in non-arthritic and arthritic glenoid bones before and after implant placement measured by digital volume correlation method, (2025), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2025.102377
14. H. Tanoto, **H. Fan**, A. Prabhu, F. Espinoza, Y. Zhou, Multi-scale investigation of orientation-dependent mechanical properties and fracture propagation in cortical bone using digital volume correlation and atomic force microscopy, (2025), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2025.102338
15. **H. Fan**, H. Tanoto, P.T. Lin, F. Bowles, S. Zhang, Z. Liu, F. Li, E. O. Klineberg, Y. Zhou, Mechanical strength across metastatic lesions in trabecular bone: The interplay of microstructure and composition, (2025), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2025.102317
16. K. Kim, S. J. Edwards, K. Fuxa, H. Lin, S. Shrestha, **H. Fan**, et al., R. Naik, Y. Zhou, X. Dong, Mucosa-Interfacing Capsule for In Situ Sensing the Elasticity of Biological Tissues, (2025), Advanced Materials Technologies, DOI: 10.1002/admt.202401487
17. H. Tanoto, Z. Zhou, K. C. Chen, R. Qiu, **H. Fan**, J. Chen, E. Milton, Y. Zhou, M. Liu, Predicting biaxial failure strengths of aortic tissues using a dispersed fiber failure model, (2025), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2024.102287
18. H. Tanoto, **H. Fan**, D. Ha, C. Ambrose, E. Klineberg, Y. Zhou, Quantifying the relation between aging-related trabecular bone microstructure and mechanical properties with digital volume correlation approach, (2024), Extreme Mechanics Letters, DOI: 10.1016/j.eml.2024.102265

19. Q. Cai, **H. Fan**, X. Li, M. Giannotta, R. Bachoo, Z. Qin, Optical modulation of the blood-brain barrier for glioblastoma treatment, (2024), Bio-protocol, DOI: 10.21769/BioProtoc.4920
20. Q. Cai, X. Li, H. Xiong, **H. Fan**, et al., E. A. Maher, R. Bachoo, Z. Qin, Optical blood-brain-tumor barrier modulation expands therapeutic options for glioblastoma treatment, (2023), Nature Communications, DOI: 10.1038/s41467-023-40579-1
21. **H. Fan**, Q. Cai, Z. Qin, Measurement and modeling of transport across the blood-brain barrier, (2023), Journal of Biomechanical Engineering, DOI: 10.1115/1.4062737
22. X. Li, Q. Cai, B. A. Wilson, **H. Fan**, H. Dave, M. Giannotta, R. Bachoo, Z. Qin, Mechanobiological modulation of blood-brain barrier permeability by laser stimulation of endothelial-targeted nanoparticles, (2023), Nanoscale, DOI: 10.1039/D2NR05062E
23. Y. Fan, C. Li, **H. Fan**, M. Yu, C. Lu, Core-Annulus radial solids concentration distribution in riser, (2018), Chemical Engineering Science, DOI: 10.1016/j.ces.2018.04.034